

Jonah Wood

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EDUCATION

Northeastern University

B.S. in Electrical and Computer Engineering

Boston, MA

Expected May 2030

Salt Lake Community College

Associate of Science

Salt Lake City, UT

Apr. 2026

EXPERIENCE

Engineering Intern

Yale Cordage

Jul. 2026 – Sep. 2026

Saco, ME

- Designed, built, budgeted, ordered, and programmed a “Smart Reeve Plate”: an end-to-end system that guides twisting-machine operators through rope reeving patterns via a scan-to-LED workflow.
- Architected the full pipeline in Node-RED, integrating database lookups, data processing and encoding, PLC and HMI programming.
- Designed and tested two custom PCBs and machined aluminum plates, and wrote Arduino firmware that receives compact varint-encoded channel data from the PLC over serial and drives their 122 addressable LEDs.
- Created a human-readable topology language and custom parser that compiles reeve patterns into hole-position and end-count data, plus authoring and editing tools built around documentation and ease of use.
- Built a microcontroller bridge that parses an industrial scale’s RS-232 output and prints weight labels on a thermal printer (TSPL), replacing a manual step.

Lead Designer & Build Team Lead

Hive 3747 – FIRST Tech Challenge Robotics

2023 – 2026

Sandy, UT

- Led the build team to back-to-back-to-back Utah State Championship Inspire Awards (1st place) and #1 robot-game rankings, qualifying for the FIRST World Championship all three seasons.
- Designed and fabricated virtually every robot mechanism, including drive trains, launchers, lifts, intakes, end-effectors, and transfers, across three years and thousands of development hours.

PROJECTS

cpp-game-engine | C++20

- 2D game engine built from scratch with SDL2/OpenGL: fixed-timestep (240 Hz) physics decoupled from rendering, SAT + AABB collision resolution, a composition-based entity/controller system, and levels authored as PNG images mapped to game objects.

Evolution Engine | Go

- Language-agnostic black-box optimizer that tunes parameters for any program over a stdin/stdout pipe, with GA and CMA-ES strategies behind a common interface and persistent worker processes for parallel evaluation.
- Reproducible, seeded runs with defaults tuned across 66 benchmark and control simulations (cart-pole, acrobot); in progress: a JSON run protocol for a live web dashboard and Bayesian optimization for expensive simulations.

TECHNICAL SKILLS

Languages: Java, C++, C (Embedded), Go, Python

Tools & Fabrication: Fusion 360 (CAD), 3D Printing, Laser Cutting, PCB Design, Node-RED, PLC/HMI, Arduino, Git, Serial (UART/RS-232)

Currently Learning: Rust, SystemVerilog

HONORS & AWARDS

National Merit Scholarship Finalist · Congressional Award Gold Medal

AP Scholar with Distinction — 8 AP exams (seven 5s, one 4), including Physics C: Mechanics, Calculus AB, and Computer Science A